

## Questions

|             |    |
|-------------|----|
| Question 1  | 2  |
| Question 2  | 3  |
| Question 3  | 4  |
| Question 4  | 5  |
| Question 5  | 6  |
| Question 6  | 7  |
| Question 7  | 8  |
| Question 8  | 9  |
| Question 9  | 10 |
| Question 10 | 11 |
| Question 11 | 12 |
| Question 12 | 13 |
| Question 13 | 14 |
| Question 14 | 15 |
| Question 15 | 16 |

1. Given that

$$y = \arctan\left(\frac{x-2}{x+2}\right)$$

where  $x \neq -2$

(a) find  $\frac{dy}{dx}$ , giving your answer in its simplest form. [2]

(b) Use integration by parts to find

$$\int \arctan\left(\frac{x-2}{x+2}\right) dx \quad [4]$$

**2.** Use integration by parts to find

$$\int_0^2 x^2 \ln(1+x) \, dx$$

giving your answer in the form  $a + b \ln 3$ , where  $a$  and  $b$  are constants to be found.

**[5]**

**3.** Use integration by parts to find

$$\int \operatorname{arsinh}(2x + 1) \, dx$$

giving your answer in terms of logarithms.

**[5]**

4.

$$h(\theta) = (3 \sin \theta - \cos \theta)(\sin \theta + 2 \cos \theta)$$

(a) Show that  $h(\theta) = \frac{1}{2} - \frac{5}{2} \cos 2\theta + \frac{5}{2} \sin 2\theta$ . [2]

(b) Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left( \frac{\pi}{2} - \theta \right) h(\theta) \, d\theta$$
 [5]

5. (a) Find

$$\int x^2 \arctan x \, dx \quad [5]$$

(b) Hence find the exact value of

$$\int_0^1 x^2 \arctan x \, dx \quad [2]$$

6. (i) Sketch the graph of  $y = \arctan x$  for  $x \in \mathbb{R}$ , indicating any horizontal asymptotes. [1]

(ii) Prove that

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2} \quad [4]$$

(iii) Using integration by parts and suitable substitutions, show that

$$\int_0^1 x(\arctan x)^2 dx = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2} \ln 2 \quad [6]$$

7. (a) Find

$$\int x^3 \arctan x \, dx \quad [5]$$

(b) Hence find the exact value of

$$\int_0^{\sqrt{3}} x^3 \arctan x \, dx \quad [2]$$

8. (a) Using the logarithmic form

$$\operatorname{artanh} x = \frac{1}{2} \ln \left( \frac{1+x}{1-x} \right), \quad |x| < 1$$

prove that the derivative of  $\operatorname{artanh} x$  is

$$\frac{1}{1-x^2}$$

[5]

(b) Hence find

$$\int_0^{1/3} \operatorname{artanh}(2x) \, dx$$

giving your answer in exact logarithmic form.

[5]

(c) Noah tries to evaluate

$$\int_0^{1/2} \operatorname{artanh}(2x) \, dx$$

using his calculator, and gets an error. Explain why.

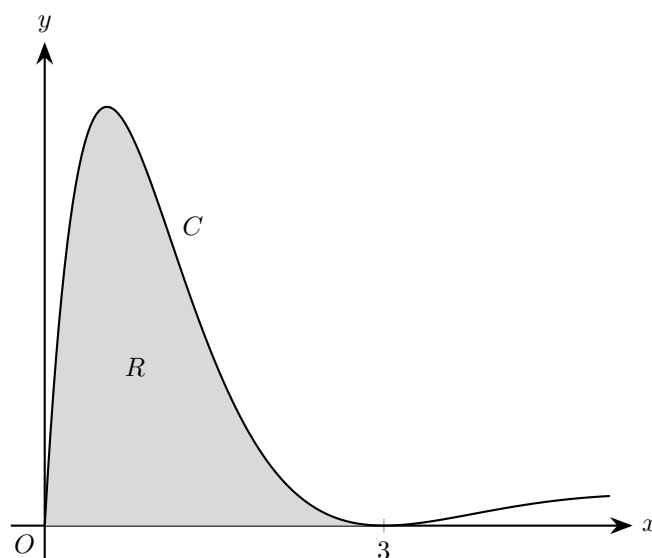
[1]

9. For  $0 < x < 1$ , use integration by parts to find

$$\int x \operatorname{artanh}(2x - 1) \, dx$$

giving your final answer in terms of logarithms.

[6]



10. The sketch shows part of the curve  $C$  given by

$$y = x(3 - x)^2 e^{-x} \quad x \geq 0$$

The shaded region  $R$  is bounded by  $C$  and the  $x$ -axis for  $0 \leq x \leq 3$ .

Find the exact area of  $R$ , giving your answer in the form

$$A + Be^{-3}$$

where  $A$  and  $B$  are rational numbers.

[5]

**11.** Show that

$$\int_{e^{-1/2}}^1 x(\ln x)^2 \, dx = a + \frac{b}{e}$$

where  $a$  and  $b$  are rational constants to be found.

**[6]**

- 12. (a)** Using the logarithmic form of  $\operatorname{artanh} x$ , prove that the derivative of  $\operatorname{artanh} x$  is

$$\frac{1}{1-x^2}$$

for  $|x| < 1$ .

[5]

- (b)** Hence find

$$\int_0^{1/2} \operatorname{artanh} x \, dx$$

giving your answer in exact logarithmic form.

[5]

- (c)** Priya tries to evaluate

$$\int_0^{3/2} \operatorname{artanh} x \, dx$$

using her calculator, and gets an ‘error’. Explain why.

[1]

**13.** Given that

$$y = \arcsin\left(\frac{x-1}{3}\right)$$

(a) find  $\frac{dy}{dx}$ , giving your answer in its simplest form. [2]

(b) Use integration by parts to find

$$\int \arcsin\left(\frac{x-1}{3}\right) dx \quad [4]$$

**14.** Use integration by parts to find

$$\int_1^9 \sqrt{x} \ln \left( \frac{9}{x} \right) dx$$

giving your answer in the form  $a + b \ln 3$ , where  $a$  and  $b$  are constants to be found.

**[5]**

**15.**

$$g(\theta) = (1 + \sin \theta - \cos \theta)^2$$

(a) Show that  $g(\theta) = 2 + 2 \sin \theta - 2 \cos \theta - \sin 2\theta$ . [2]

(b) Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \theta g(\theta) \, d\theta$$
 [5]